Multi-Agent Framework for Healthcare Fraud Detection

The financial integrity of the U.S. healthcare system is continuously undermined by fraudulent activities, waste, and abuse (FWA). Industry estimates indicate that healthcare fraud accounts for $60 billion to $68 billion in annual losses, representing 3% to 10% of total healthcare expenditures. Traditional detection mechanisms typically rely on static rule-based engines or highly retroactive human audits, which often result in a “pay and chase” dilemma where recovering disbursed funds is resource-intensive and leads to high provider abrasion. This report explores the application of Agentic Artificial Intelligence, specifically through a Multi-Agent System (MAS) architecture, for healthcare payment integrity and FWA detection. In this MAS framework designed for healthcare, specialized AI agents handle discrete aspects of the claims lifecycle. Instead of relying on a classical ML model, domain-specific agents such as deterministic coding validators, generative medical necessity checkers, and unsupervised anomaly detectors communicate to execute claim reviews.

Trends in Fraud, Waste and Abuse Detection

Existing work in fraud detection in healthcare claims uses machine learning, in a domain where an estimated 3 to 10 percent of expenditure is lost to fraud, waste, and abuse [2]. Across 137 studies, supervised classifiers account for the majority of applications, with ensemble tree methods reporting the highest area under the curve, while unsupervised outlier methods such as Local Outlier Factor and Isolation Forest operate where labels are absent [1]. However, the paradigm of relying on single-model classifiers and rule-based edits faces structural limitations. Single-model classifiers optimized for recall produce high volumes of false positives, and without adversarial validation, these errors propagate to human investigators [1 , 3]. Furthermore, these models operate purely on individual claims or isolated provider profiles [4]. While Large Language Models (LLMs) offer a potential solution for unstructured data, deploying general-purpose LLMs without domain-specific guardrails introduces risks of data leakage and hallucination [5].

Multi-agent architectures directly address these constraints by shifting the paradigm from static scoring to assisting dynamic decision-making. Unlike traditional supervised models that depend on fixed numerical searches and output risk scores, agentic frameworks can mirror the decision making process through distinct, coordinated roles. For example, where a traditional claim-level classifier might flag a physician simply for high billing volume, an agentic system can autonomously orchestrate graph-based analytics to map the underlying topology of a network and expose illicit routing between physicians and suppliers [6]. Crucially, agents introduce the capacity for automated adversarial validation; a "critique" agent can actively stress-test initial detections by generating and investigating plausible benign counter-hypotheses (e.g., verifying if a flagged anomaly actually represents a legitimate multi-state telemedicine practice). By synthesizing this validated evidence into explicit confidence tiers with traceable reasoning chains, multi-agent frameworks filter out false positives before they reach investigators, bridging the gap between raw statistical outliers and actionable operational intelligence.

Within these agentic systems, the method of information retrieval dictates the accuracy of clinical and policy determinations. Standard retrieval augmented generation is constrained for this function because it returns unstructured text segments that do not supply the structured reasoning required for clinical decisions. Graph based retrieval organizes retrieved content as a knowledge graph and supports reasoning across related entities rather than isolated passages [7]. In the medical domain, MedGraphRAG applies this method to produce responses grounded in evidence with explicit source citations and reports higher accuracy than prior models across medical question answering benchmarks[8]. The same structure transfers to payment integrity: encoding procedures, diagnoses, and policies as a graph anchors retrieval to the policy that governs a service, and temporal edges restrict retrieval to the policy version in force on the date of service, which yields determinations that cite the applicable rule and its effective period.

Opportunities
Proactive Prepay Intervention: Implementing a multi-agent system shifts the paradigm from retroactive "pay and chase" methodologies to dynamic prepay prevention. By utilizing specialized domain agents to automate complex clinical reasoning, improper payments can be captured earlier in the claims lifecycle.

Elimination of Time-Drift Errors: Leveraging temporal GraphRAG ensures that medical determinations are strictly anchored to the specific payer policies active on a claim's date of service. This deterministic temporal filtering effectively eliminates the hallucination risks associated with mid-year code retirements and policy versioning.

Optimized Investigator Efficiency: Introducing a budget-constrained adjudication agent enables human-in-the-loop (HITL) workflows to be prioritized by return on investment (ROI) [10]. By synthesizing evidence into explicit confidence tiers and generating prioritized worklists, the administrative burden on Special Investigative Units (SIUs) is significantly reduced.

Threats
Regulatory Scrutiny of "Black Box" AI: In a highly regulated healthcare environment, generating automated financial denials without traceable, explicit reasoning chains risks severe provider abrasion, costly appeals, and compliance penalties.

Temporal and Semantic Hallucinations: Utilizing naive Retrieval-Augmented Generation (RAG) pipelines without date-of-service guardrails can cause the AI to confidently apply temporally expired payer policies, resulting in legally invalid claim denials.

Model and Agent Drift: The internal logic and prompting efficacies of autonomous agents can steadily degrade as they encounter unhandled edge cases and evolving fraud typologies, which requires continuous adversarial validation and self-reflective adaptation.

Strategic Options for Cotiviti to Explore

To capitalize on agentic AI for Treatment, Payment, and Operations (TPO) while maximizing SIU efficiency, Cotiviti can explore the strategic deployment of specialized, coordinated AI agents.

Option 1: Scale Temporal GraphRAG for Medical Necessity Reviews
Cotiviti can build upon the proof-of-concept's temporal GraphRAG architecture to construct a comprehensive knowledge-graph layer with learned entity and relation embeddings [9]. By mapping unstructured payer policies to clinical ontologies and utilizing a claim's date of service as a hard filter, this agent guarantees that clinical determinations are highly accurate, explicitly cited, and temporally valid.

Option 2: Deploy Prior-Guided Claim Routing
To optimize computational resources and minimize provider abrasion, Cotiviti can explore implementing a prior-routing orchestrator agent. By utilizing Bayesian posteriors informed by historical datasets such as OIG-LEIE exclusion overrides and facility-network terms this orchestrator can intelligently route incoming claims to either a fast-track deterministic coding check or a deep multi-agent evaluation.

Option 3: Implement Unsupervised Peer-Baseline Profiling
To address the scarcity of labeled fraud data, Cotiviti could invest in unsupervised anomaly detection agents. By leveraging massive open datasets like the CMS Part B Provider Utilization API to establish legitimate peer baselines, this agent can calculate specialty-specific z-scores to flag statistical outlier utilization volumes and identify emerging fraud schemes.

Acknowledgements

This report was written using the assistance of AI (Gemini) for grammar and paraphrasing.

References

1. du Preez, A., Bhattacharya, S., Beling, P., & Bowen, E. (2025). Fraud detection in healthcare claims using machine learning: A systematic review. *Artificial Intelligence in Medicine*, *160*, 103061.
2. Leveraging AI to Help Detect Fraud in Medical Claims | cloud-infrastructure - Oracle Blogs, June 29, 2026 https://blogs.oracle.com/cloud-infrastructure/leveraging-ai-help-detect-fraud-in-medical-claims
3. Smith, S. R., Wright, H. R., & Moore, J. L. Explainable Predictive Analytics for Fraud, Resource Allocation, and Security in US Healthcare Systems.
4. R. A. Bauder and T. M. Khoshgoftaar, "Medicare Fraud Detection Using Machine Learning Methods," *2017 16th IEEE International Conference on Machine Learning and Applications (ICMLA)*, Cancun, Mexico, 2017, pp. 858-865, doi: 10.1109/ICMLA.2017.00-48.
5. Jung K. H. (2025). Large Language Models in Medicine: Clinical Applications, Technical Challenges, and Ethical Considerations. Healthcare informatics research, 31(2), 114–124. https://doi.org/10.4258/hir.2025.31.2.114
6. Ahmed Bin Osman, Ayomide owolabi. Generative and Agentic Artificial Intelligence for Medical Coding and Billing: A Human-in-the-Loop Architecture and Evaluation, 13 February 2026, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-8863592/v1]
7. Han, H., Wang, Y., Shomer, H., Guo, K., Ding, J., Lei, Y., ... & Tang, J. (2024). Retrieval-augmented generation with graphs (graphrag). *arXiv preprint arXiv:2501.00309*.
8. Wu, J., Zhu, J., Qi, Y., Chen, J., Xu, M., Menolascina, F., ... & Grau, V. (2025, July). Medical graph rag: Evidence-based medical large language model via graph retrieval-augmented generation. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 28443-28467).
9. Hu, Q., Tu, X., Guo, C., & Zhang, S. (2025, April). Time-aware ReAct agent for temporal knowledge graph question answering. In *Findings of the Association for Computational Linguistics: NAACL 2025* (pp. 6028-6039).
10. He, Y., Li, R., Chen, A., Liu, Y., Chen, Y., Sui, Y., ... & Hooi, B. (2025, November). Enabling self-improving agents to learn at test time with human-in-the-loop guidance. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track* (pp. 1625-1653).